NEXVIA AEGIS V9-M4 Independent Statistical Audit

Phase 21 Report: Source-of-Truth Protocol Consistency Audit

Reconciliation of Decision Rules, Estimands, and Holdout Allocations in Report 20

Autonomous Verification Subsystem (Google DeepMind Antigravity)

Git Branch: v9.0-runtime-assurance · *Starting Commit: a8895a0* · *Date: October 26, 2026*

CRITICAL NON-AUTHORIZATION DIRECTIVE:
This audit resolves statistical decision thresholds, estimands, sample size partitions, and scenario allocations before the project owner makes an authorization decision for Milestone V9-M4 (Confirmatory Holdout Evaluation). **This document does NOT constitute authorization to execute Milestone V9-M4, nor does it provide or invoke any execution token.** All evaluations were conducted strictly using synthetic test fixtures, in-memory models, and read-only inspection of existing calibration and validation artifacts. Production holdout manifests (`test_mix_manifest_r1.json` and `test_scen_manifest_r1.json`) and output directories (`results/v9/test_mix/` and `results/v9/test_scen/`) do not exist. Reserved holdout seed blocks (73000..74439 for mixture holdout and 75000..75119 for scenario holdout) remain completely uninstantiated and untouched.

Abstract

This audit report provides a rigorous source-of-truth reconciliation of Report 20 (`20_v9_m4_owner_authorization_readines`) against the governing preregistered protocol (`docs/v9_experiment_protocol.md`), the production holdout harness (`experiments/v9_evaluate_holdout.py`), and the synthetic regression test suite (`tests/test_v9_holdout_preflight`). The review investigates potential discrepancies regarding Hypothesis H_1 selective risk thresholds, Hypothesis H_4 coverage floor preconditions, Scenario Holdout total episode counts, Hypothesis H_5 scenario allocations, and Hypothesis H_3 sample size scope.

The audit establishes that: (1) the production implementation and the authoritative preregistered protocol are in **100% perfect agreement**; (2) all identified discrepancies in Report 20 are **purely narrative and transcription errors** in informal conversational summaries and descriptive text; (3) the authoritative threshold for H_1 is $U_{\mathrm{CP},1} \leq 0.05$ on $M \geq 59$ (not 0.01); (4) the authoritative threshold for H_4 is sample coverage $\hat{C} \geq 0.35$ (not 0.95); (5) the Scenario Holdout partition strictly comprises 12 scenarios with 120 episodes each ($N = 1,440$, seeds 75000..75119 under branch $p = 5$); (6) Hypothesis H_5 evaluates across all 12 scenarios ($N_s = 120$ each); and (7) $N = 1,200$ is exclusively the matched-pair count of the S_{10} subset evaluated under H_3. Zero code defects exist. All 62 preflight and 173 full regression tests pass. The formal status is declared as **M4_REPORTING_CORRECTIONS_REQUIRED**.

1 Executive Summary

1.1 Executive Summary (English)

1. **Status Determination: M4_REPORTING_CORRECTIONS_REQUIRED.** The evaluation code and preregistered protocol are mathematically and algorithmically sound and match 100%. However, formal reporting corrections are required to supersede clerical transcription errors in Report 20.
2. **Hypothesis H_1 Threshold (0.05 vs 0.01):** The registered and implemented threshold is $U_{\mathrm{CP},1}(K, M; 0.05) \leq 0.05$ on $M \geq 59$. The mention of 0.01 in Report 20's summary table was an informal transcription conflation with the M2R2 calibration loss-to-tolerance parameter $\alpha_{\mathrm{mon}} = 0.010$.
3. **Hypothesis H_4 Precondition (0.35 vs 0.95):** The registered and implemented threshold is empirical sample coverage $\hat{C} = M/N \geq 0.35$ ($M \geq 504$ on $N = 1,440$). The mention of 0.95 in Report 20's summary table conflated task decision coverage with statistical confidence level $1 - \delta = 0.95$ or conformal set coverage.
4. **Scenario Holdout Sampling Design ($1,440$ vs $1,200$):** The Scenario Holdout partition comprises 12 scenarios $\times$ 120 seeds/scenario $= 1,440$ episodes. The number $1,200$ applies exclusively to the 10-scenario core block (S_{10}) evaluated for H_3 TSR noninferiority.
5. **Hypothesis H_5 Scenario Allocation:** Evaluates across all 12 scenarios with $N_s = 120$ episodes per scenario. Erroneous narrative mentions of "6 scenarios with $N = 200$" or "$N_s = 100$" were spurious arithmetic factorizations of $1,200$.

1.2 บทสรุปสำหรับผู้บริหาร (Executive Summary in Thai)

1. **ผลการตรวจสอบความสอดคล้อง:** `M4_REPORTING_CORRECTIONS_REQUIRED` รหัสต้นฉบับการประเมินผลและพิธีสารที่จดทะเบียนไ ไม่พบข้อบกพร่องในโค้ดหรือการกำหนดค่าทางสถิติ แต่จำเป็นต้องบันทึกข้อแก้ไขเชิงรายงานเพื่อแก้ไขข้อความคลาดเคลื่อนใน Report 20
2. **สมมติฐาน H_1 (Selective Risk):** เกณฑ์ที่ถูกต้องคือ $U_{\mathrm{CP},1} \leq 0.05$ บนจำนวนการตัดสินใจ $M \geq 59$ (การระบุ 0.01 เกิดจากการสับสนกับค่าขอบเขตความคลาดเคลื่อนในการสอบเทียบ $\alpha_{\mathsf{mon}} = 0.010$)
3. **สมมติฐาน H_4 (Task Coverage):** เกณฑ์สัดส่วนการตัดสินใจขั้นต่ำที่ถูกต้องคือ $\hat{C} \geq 0.35$ หรือ 35% ($M \geq 504$) (การระบุ 0.95 เกิดจากการสับสนกับระดับความเชื่อมั่น $1 - \delta = 0.95$)
4. **การจัดสรรตัวอย่าง Scenario Holdout และ H_5:** การออกแบบที่แท้จริงคือ 12 ฉากทัศน์ × 120 ตอน = 1,440 ตอน สำหรับขนาด $N = 1,200$ นั้นเป็นขนาดตัวอย่างเฉพาะกลุ่มฉากทัศน์หลัก 10 ฉากทัศน์แรก (S_{10}) ที่ใช้ทดสอบความไม่ด้อยกว่า (H_3) เท่านั้น

2 Repository Identity and Evidence Inspected

- **Current HEAD Commit:** `a8895a0` on branch `v9.0-runtime-assurance`.
- **Remote Sync Status:** Exactly synchronized with `origin/v9.0-runtime-assurance`.
- **Report 20 Markdown SHA-256:** `36502a1207e316c3a5b843f92370400a150f1195f2709200456353eabdd5d9f0`.
- **Report 20 PDF SHA-256:** `6a38063c7cd4af5e33ad93ba0063abd630ff4b438d32dc327657543c9ce64443`.
- **Authoritative Protocol Document:** `docs/v9_experiment_protocol.md` (Step 3 of 3).
- **Evaluator Implementation:** `experiments/v9_evaluate_holdout.py`.
- **Synthetic Preflight Suite:** `tests/test_v9_holdout_preflight.py`.

3 Discrepancy-by-Discrepancy Source-of-Truth Reconciliation

Dimension	Report 20 Narrative	Registered Protocol	Production Code	Resolution Status
H_1 Bound	UCB $\leq$ 0.01 (summary)	$U_{\mathrm{CP},1} \leq 0.05$ on $M \geq 59$	$U_{\mathrm{CP},1} \leq 0.05$ on $m \geq 59$	Transcription Error in summary table. Authoritative: 0.05.
H_4 Coverage	$\hat{C} \geq 0.95$ (summary)	Point estimate $\hat{C} \geq 0.35$	Point estimate $\hat{C} \geq 0.35$	Transcription Error in summary table. Authoritative: $\geq$ 0.35.
Scenario Total	$N = 1,200$ (summary)	12 scen × 120 seeds = 1,440	$N_{\mathrm{scen}} = 1,440$ fail-closed	Conflation Error with S_{10}. Total is 1,440.
H_5 Allocation	"6 scen × 200" / "$N_s = 100$"	12 scen × 120 eps ($N_s = 120$)	12 scen, $n_s = 120$ checked	Spurious factorizations of 1,200. True: $12 \times 120 = 1,440$.
H_3 Scope	Conflated with total	S_{10} block ($10 \times 120 = 1,200$)	Filtered to S_{10} (1,200 pairs)	Reconciled: $N = 1,200$ is exclusively S_{10} subset.

Table 1: Comprehensive Source-of-Truth Reconciliation across Protocol, Code, and Reports.

4 Detailed Protocol and Code Alignment

4.1 Hypothesis H1 (Selective Risk Certification)

- **Mathematical Formulation:** $\mathcal{H}_{0,1} : C = 0 \vee R_{\text{sel}} > 0.05$ vs. $\mathcal{H}_{1,1} : C > 0 \wedge R_{\text{sel}} \leq 0.05$.
- **Decision Rule:** Evaluated on $\mathcal{D}_{\text{test}}^{\text{mix}}$ ($N = 1,440$). Requires $M \geq 59$ and exact one-sided 95% Clopper–Pearson UCB $U_{\text{CP},1}(K, M; 0.05) \leq 0.05$.
- **Boundary Behavior:** When $M = 0$, R_{sel} evaluates as `"UNDEFINED"`, $U_{\text{CP},1} = \text{NaN}$, and H_1 evaluates to `passed = False` with status `"INSUFFICIENT_COVERAGE"`.

4.2 Hypothesis H4 (Task Coverage Floor Precondition)

- **Decision Rule:** $\hat{C} = M/N \geq 0.35$ ($M \geq 504$ on $N = 1,440$).
- **Anti-Gaming Function:** Precludes trivial risk minimization via excessive abstention. When $\hat{C} < 0.35$, the evaluator classifies the policy as `"CONSERVATIVE_RISK_COVERAGE_TRADEOFF"`.

4.3 Scenario Holdout Sample Size and Allocation Table

Partition / Block	Scenarios	Seeds/Scen	Total Episodes	Matched Pairs	Evaluated Hypotheses
Mixture Holdout ($\mathcal{D}_{\text{test}}^{\text{mix}}$)	Uniform Mixture	N/A	1,440	1,440	H_1 (Risk), H_2 (Incidence), H_4 (Coverage)
Scenario Holdout ($\mathcal{D}_{\text{test}}^{\text{scen}}$)	12 ($S_{10} \cup S_{\text{ext}}$)	120	1,440	1,440	Full Stratified Grid Suite
– S_{10} Core Subsample	10 (Scenarios 1–10)	120	1,200	1,200	H_3 (TSR Noninferiority vs V8.1 & V7)
– S_{ext} Stress Subsample	2 (Scenarios 11–12)	120	240	240	Stress Breakdown Diagnostic
All 12 Scenarios	12 (Full Grid)	120	1,440	N/A	H_5 (Berger IUT & Holm–Šidák Step-Down)

Table 2: Reconciled Sampling Allocation and Hypothesis Mapping for Confirmatory Holdout.

4.4 Hypothesis H5 (Worst-Case Scenario Safety Envelope)

- **Global Conjunction (Berger IUT):** Unconditional false-success incidence null $H_{0,5,s} : R_{\text{inc},s} > 0.05$. Component p-value $p_{\text{inc},s} = \text{binom. cdf}(\text{FS}_s, N_s = 120, 0.05)$. Rejection rule: $\max_{s \in \mathcal{S}_{12}} p_{\text{inc},s} \leq 0.05$.
- **Auxiliary Multiplicity Adjustment:** Holm–Šidák sequential step-down stopping rule $k^* = \min\{j : p_{(j)} > \alpha_{(j)}\}$ controls FWER for individual scenario claims.
- **Zero Observed False Successes:** At $\text{FS}_s = 0$, $p = 0.95^{120} \approx 0.002055 \leq 0.05$. Confirms safety without falsely claiming zero underlying physical risk.

4.5 Hypothesis H3 (TSR Noninferiority Scope)

- **Domain:** Scenario block S_{10} ($N = 1,200$ matched pairs).
- **Controlling Rule:** Student's t Wald LCB (> -0.05) and nominal TSR ≥ 0.94.
- **Sensitivity:** Newcombe Method 10 reported concurrently as sensitivity evidence.

5 Regression Testing and Holdout Isolation

- Focused Preflight Suite: **62/62 passed** in 1.44s (Exit code: `0`).
- Full Repository Suite: **173/173 passed** in 31.44s (Exit code: `0`).
- Holdout Isolation: Production manifests (`test_mix_manifest_r1.json`, `test_scen_manifest_r1.json`) and directories (`results/v9/test_mix/`, `results/v9/test_scen/`) confirmed absent. Reserved seed blocks (73000..74439 and 75000..75119) untouched.

6 Final Determination

M4_REPORTING_CORRECTIONS_REQUIRED

All statistical implementation and test code agree with the preregistered protocol. Reporting errata in Report 20 are formally superseded by this Report 21.

EXPLICIT NON-AUTHORIZATION STATEMENT:
This consistency audit does NOT authorize execution of Milestone V9-M4. The production evaluator remains locked pending explicit owner authorization.